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DEPARTMENT OF CSE

PROJECT ABSTRACT SUBMISSION

COURSE TITLE & CODE : DATA SCIENCE (24AD3107) A.Y. 2026 - 2027

Title of the project	Handwritten Text Recognition using Deep Learning	
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Abstract

Handwritten text recognition is an important application of deep learning and computer vision that enables computers to recognize and convert handwritten characters into digital text. This project focuses on developing a deep learning model using the EMNIST (Extended Modified National Institute of Standards and Technology) dataset, which contains handwritten letters and digits. The model is trained on labelled handwritten character images and learns important visual patterns such as strokes, edges, shapes, and character structures. A Convolutional Neural Network (CNN) can be used to automatically extract relevant features from the images and classify handwritten characters into their respective categories. The project includes data preprocessing, image normalization, model development, training, validation, performance evaluation, and prediction. Evaluation metrics such as accuracy, loss, precision, recall, and confusion matrix can be used to measure the performance of the model. The trained model can then be used as a basic handwritten text recognition system by processing handwritten character images and converting the recognized characters into digital text. The project demonstrates how deep learning and computer vision techniques can be applied to handwritten document analysis and automated text recognition.